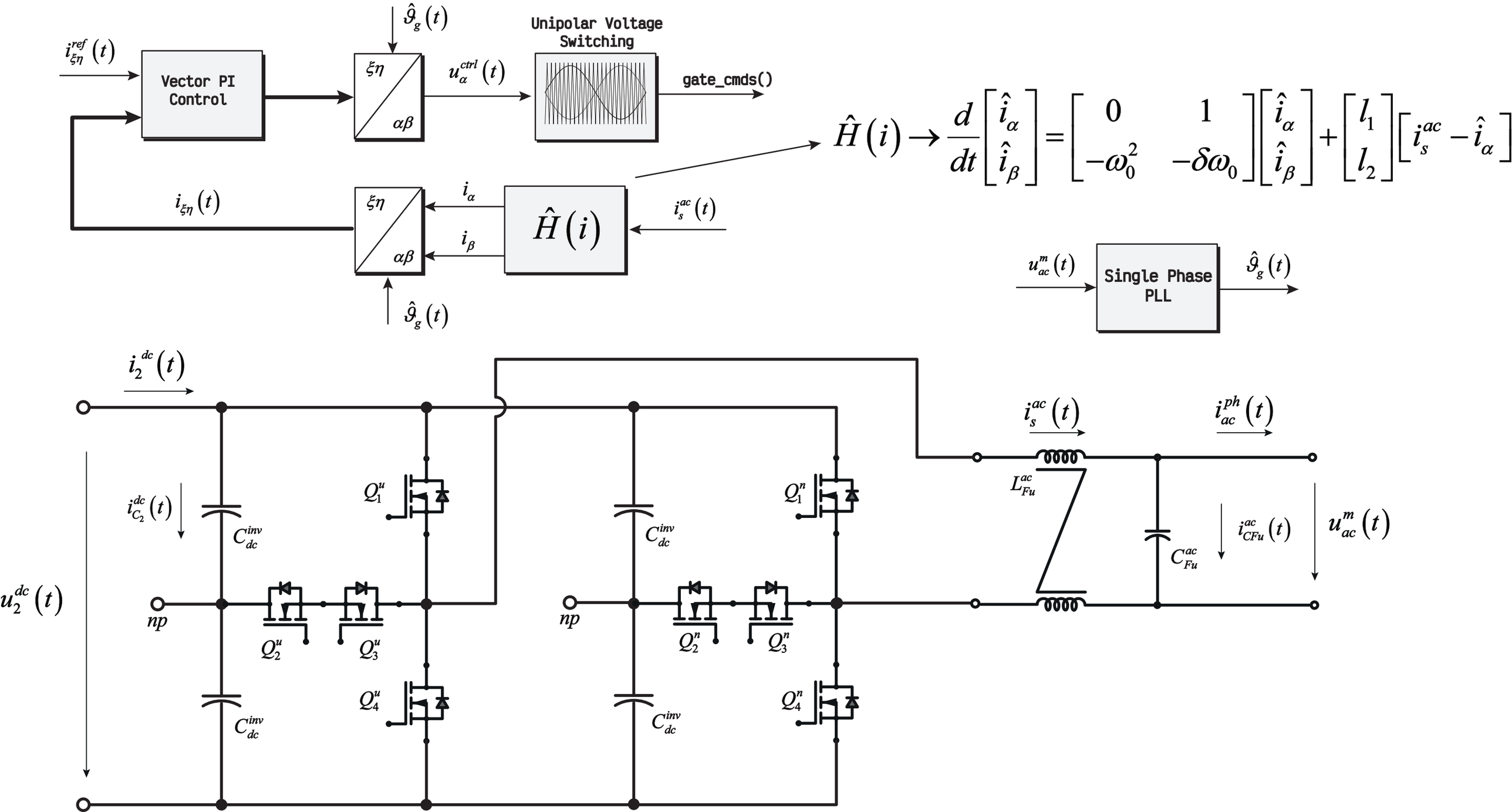

$i_{\xi\eta}^{ref}(t)$
Vector PI Control
$\hat{\vartheta}_g(t)$
$\xi\eta$
$\alpha\beta$
$u_\alpha^{ctrl}(t)$
Unipolar Voltage Switching
gate_cmds()
$i_{\xi\eta}(t)$
i_α
i_β
$\hat{H}(i)$
$i_s^{ac}(t)$
$\hat{H}(i) \rightarrow \frac{d}{dt}\begin{bmatrix}\hat{i}_\alpha \\ \hat{i}_\beta\end{bmatrix} = \begin{bmatrix}0 & 1 \\ -\omega_0^2 & -\delta\omega_0\end{bmatrix}\begin{bmatrix}\hat{i}_\alpha \\ \hat{i}_\beta\end{bmatrix} + \begin{bmatrix}l_1 \\ l_2\end{bmatrix}\left[i_s^{ac} - \hat{i}_\alpha\right]$
$u_{ac}^m(t)$
Single Phase PLL
$i_2^{dc}(t)$
$i_{C_2}^{dc}(t)$
C_{dc}^{inv}
Q_1^u
Q_2^u
Q_3^u
Q_4^u
Q_1^n
Q_2^n
Q_3^n
Q_4^n
np
$u_2^{dc}(t)$
$i_s^{ac}(t)$
$i_{ac}^{ph}(t)$
L_{Fu}^{ac}
C_{Fu}^{ac}
$i_{CFu}^{ac}(t)$
$u_{ac}^m(t)$